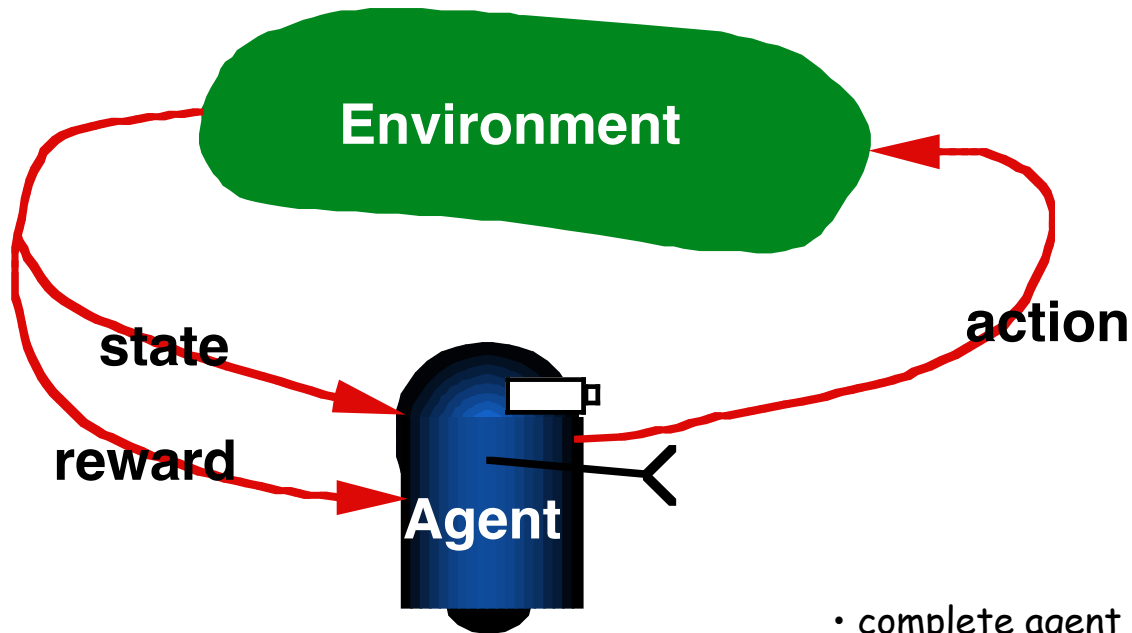


Examples and Videos of Markov Decision Processes (MDPs) and Reinforcement Learning

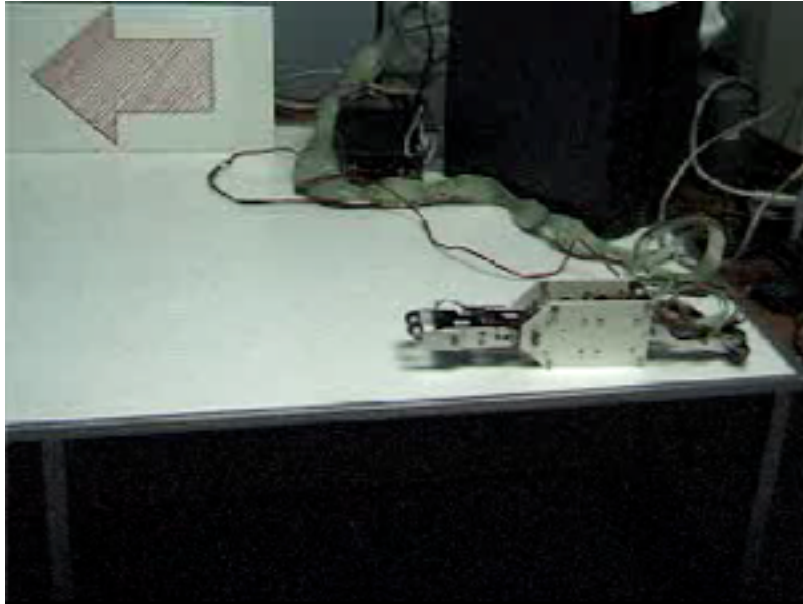
Artificial Intelligence is
interaction to achieve a goal



- complete agent
- temporally situated
- continual learning & planning
- object is to affect environment
- environment stochastic & uncertain

States, Actions, and Rewards

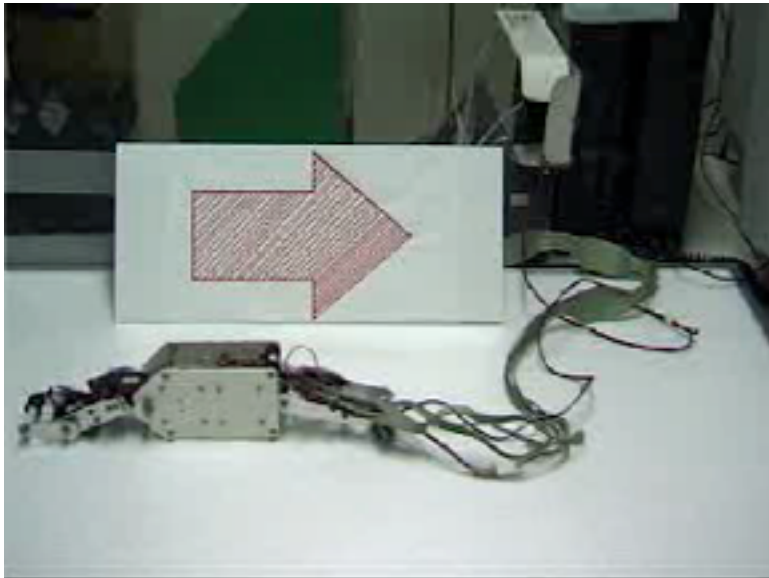
Hajime Kimura's RL Robots



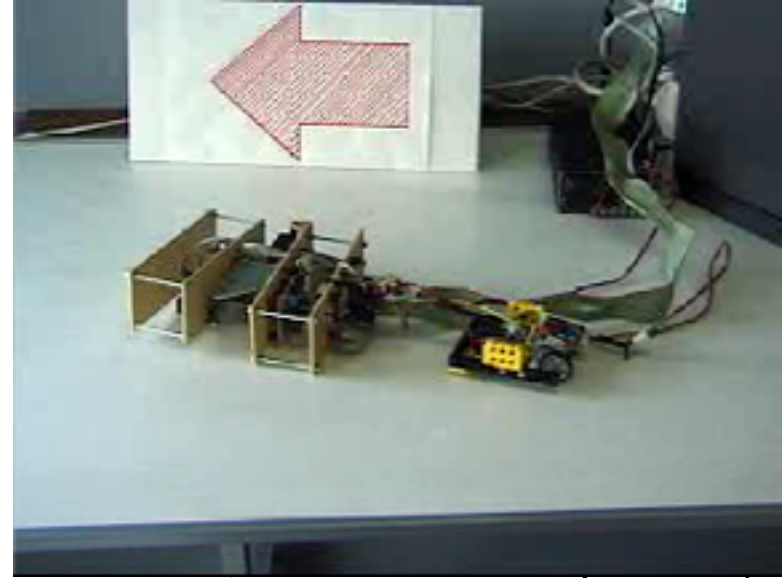
Before



After

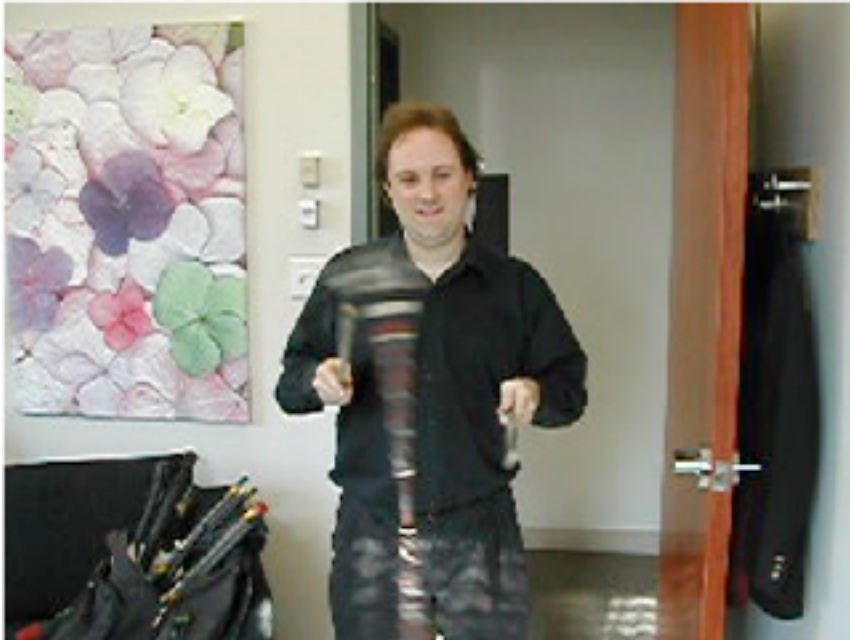


Backward

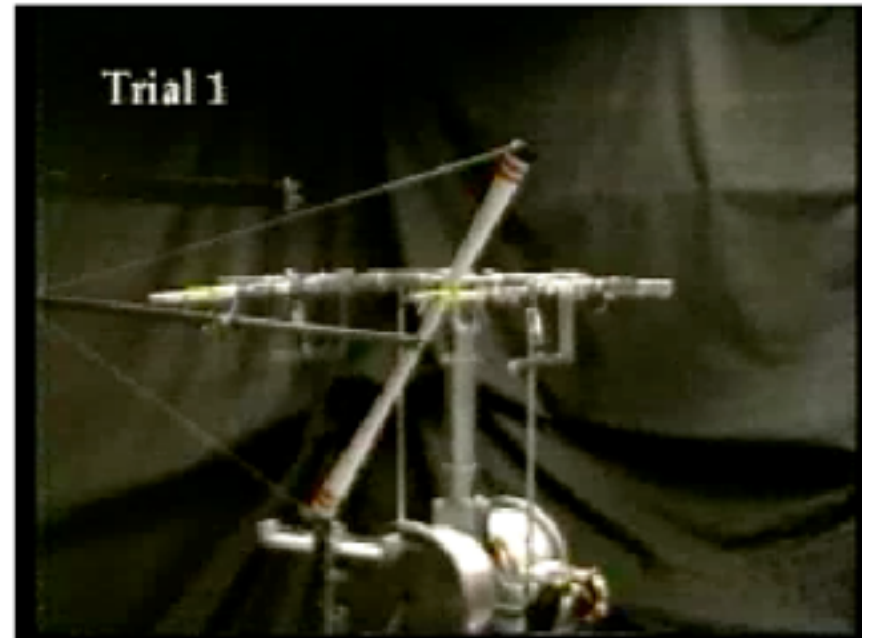


New Robot, Same algorithm

Devilsticking



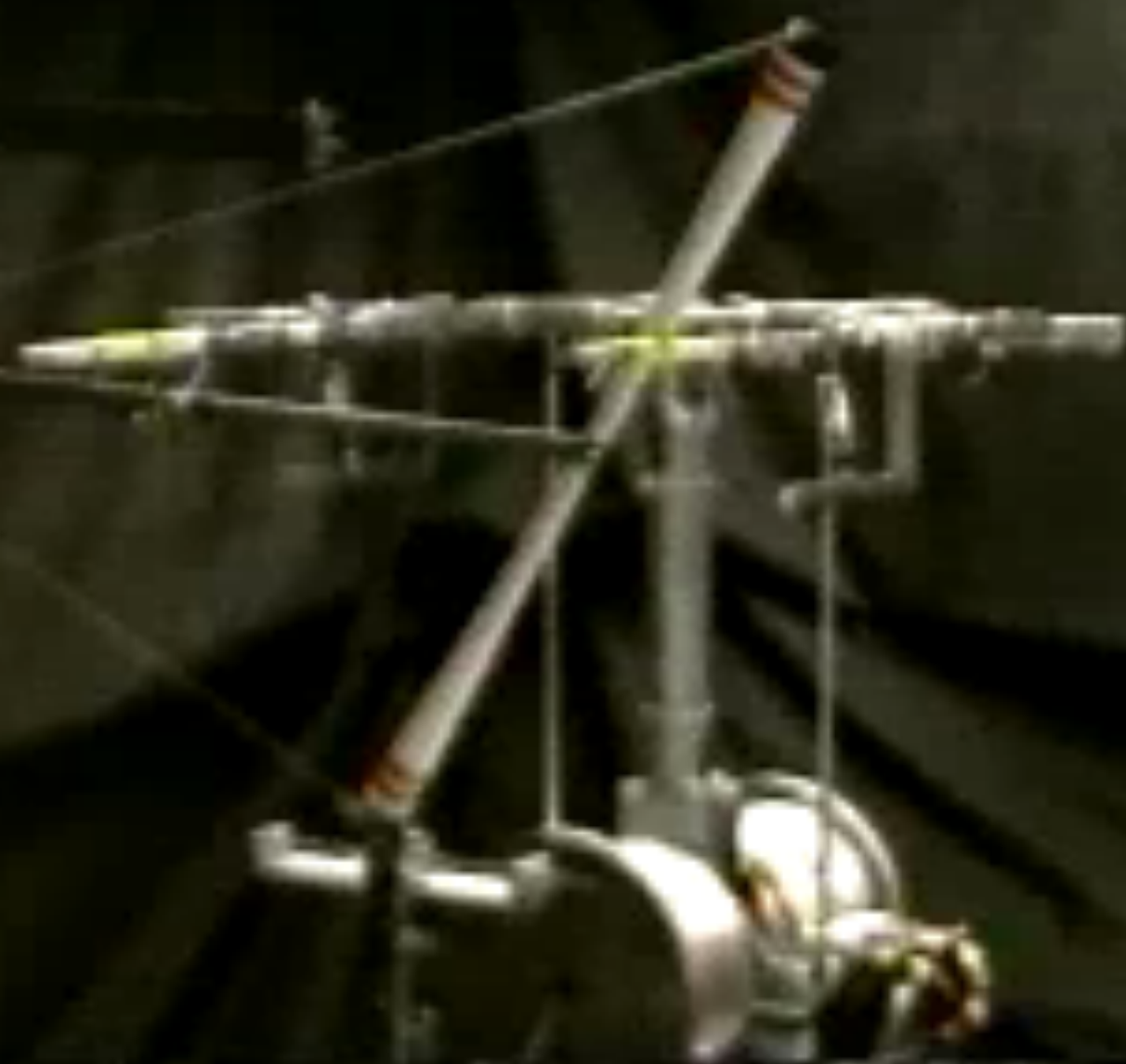
Finnegan Southey
University of Alberta



Stefan Schaal & Chris Atkeson
Univ. of Southern California
“Model-based Reinforcement
Learning of Devilsticking”



Trial 1



The RoboCup Soccer Competition



Autonomous Learning of Efficient Gait

Kohl & Stone (UTexas) 2004









Policies

- A **policy** maps each state to an action to take
 - Like a stimulus–response rule
- We seek a policy that maximizes cumulative reward
- The policy is a subgoal to achieving reward

The Reward Hypothesis

The goal of intelligence is to maximize the cumulative sum of a single received number:

“reward” = pleasure - pain

Artificial Intelligence = reward maximization

Value

Value systems are hedonism with foresight

We value situations according to how much reward we expect will follow them

All efficient methods for solving sequential decision problems determine (learn or compute) “value functions” as an intermediate step

Value systems are a *means* to reward, yet we *care more* about values than rewards

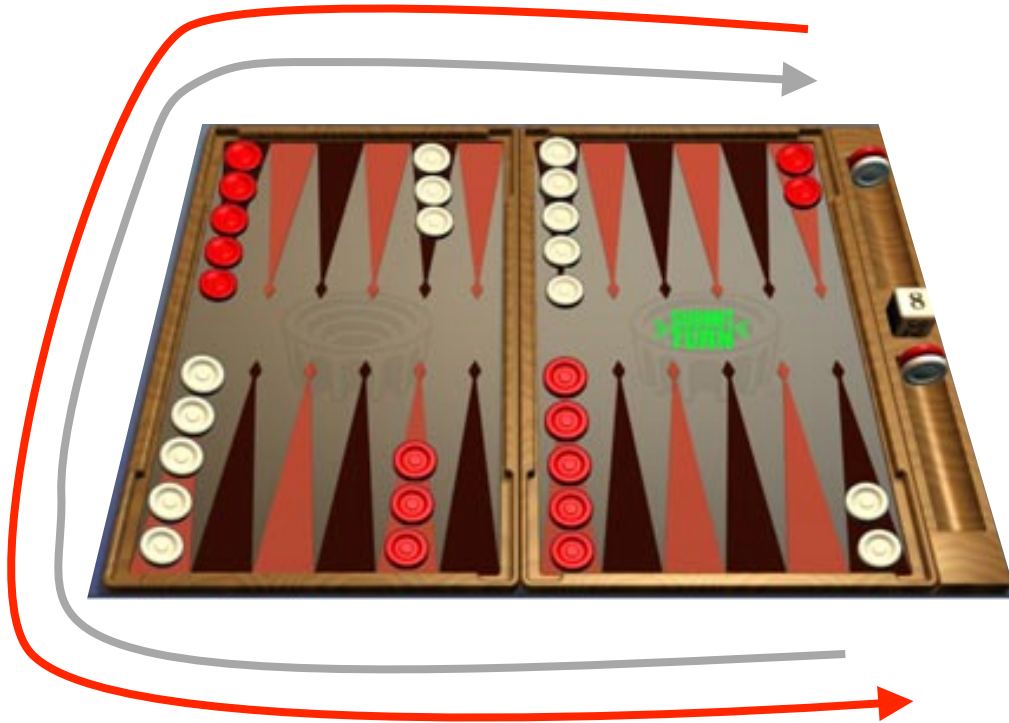
Pleasure = Immediate Reward
≠ good = Long-term Reward

“Even enjoying yourself you call evil whenever it leads to the loss of a pleasure greater than its own, or lays up pains that outweigh its pleasures. ... Isn't it the same when we turn back to pain? To suffer pain you call good when it either rids us of greater pains than its own or leads to pleasures that outweigh them.”

—Plato, Protagoras



Backgammon



STATES: configurations of the
playing board ($\approx 10^{20}$)

ACTIONS: moves

REWARDS: win: +1

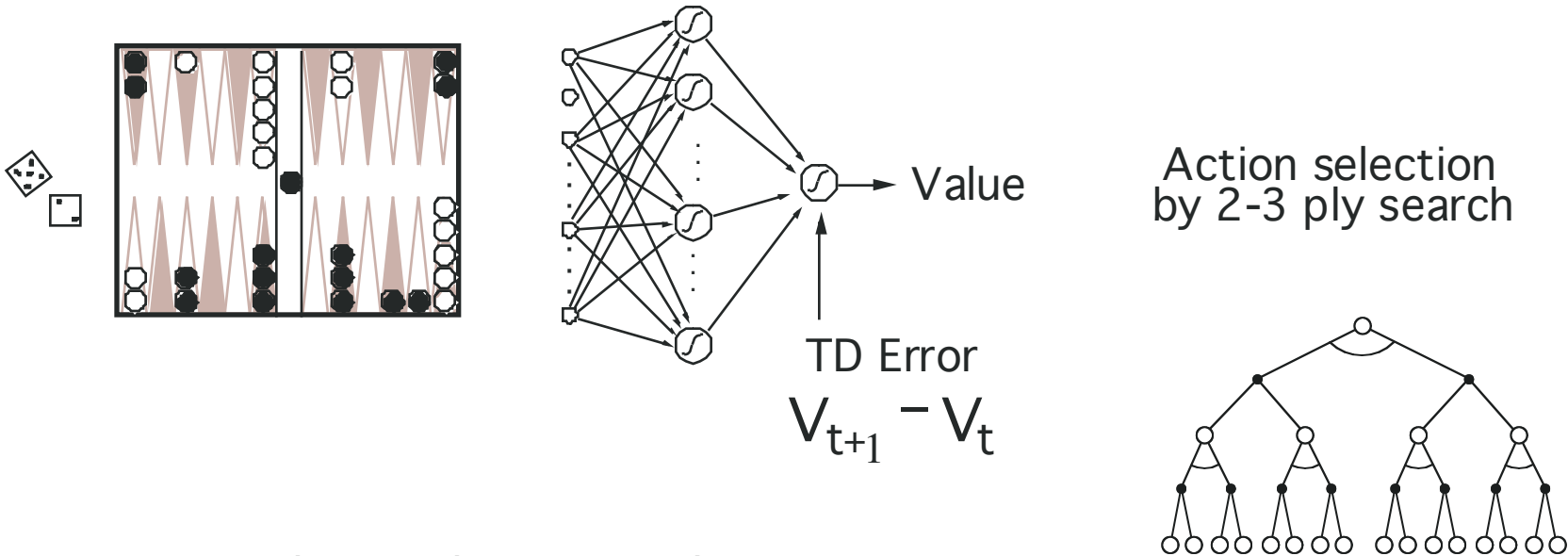
lose: -1

else: 0

a “big” game

TD-Gammon

Tesauro, 1992-1995



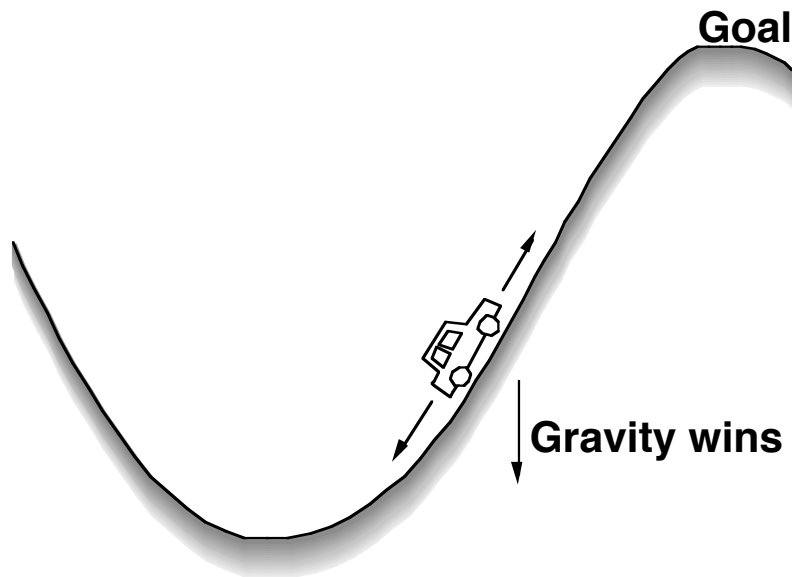
Start with a random Network

Play millions of games against itself

Learn a value function from this simulated experience

Six weeks later it's the best player of backgammon in the world

The Mountain Car Problem



SITUATIONS: car's position and velocity

ACTIONS: three thrusts: forward, reverse, none

REWARDS: always -1 until car reaches the goal

No Discounting

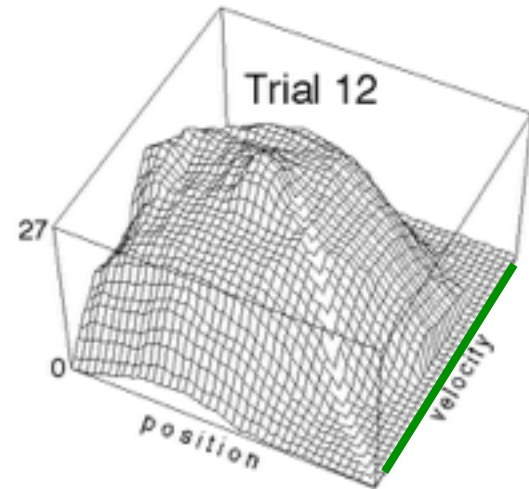
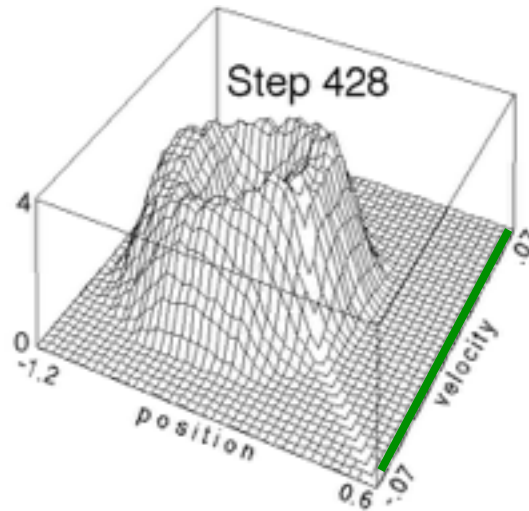
Minimum-Time-to-Goal Problem

Value Functions Learned while solving the Mountain Car problem

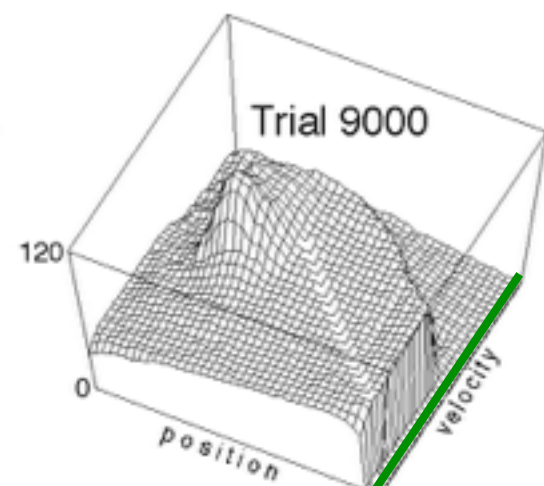
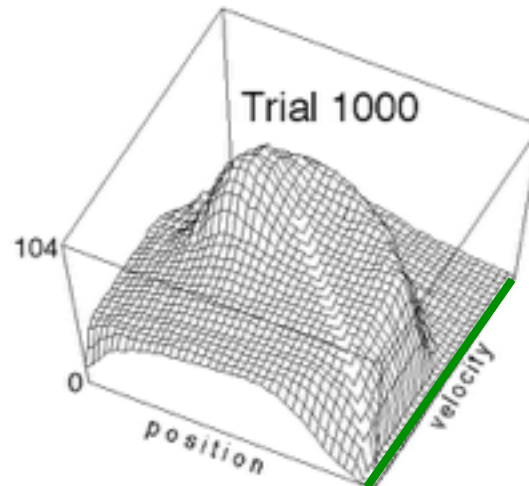
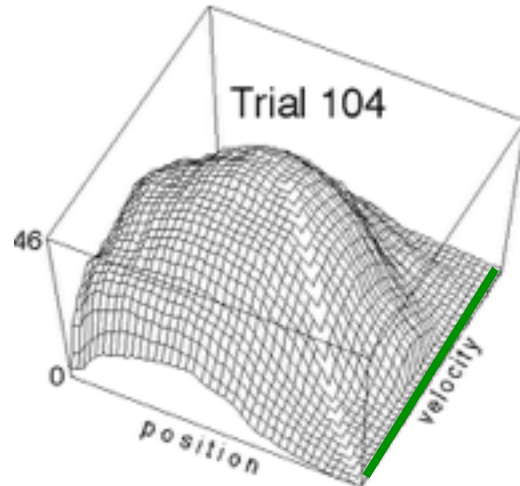


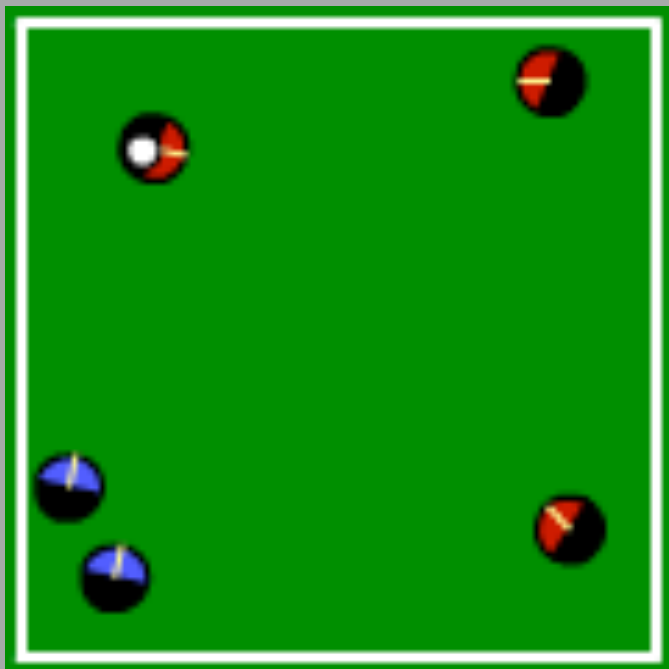
Minimize Time-to-Goal

Value = estimated time to goal

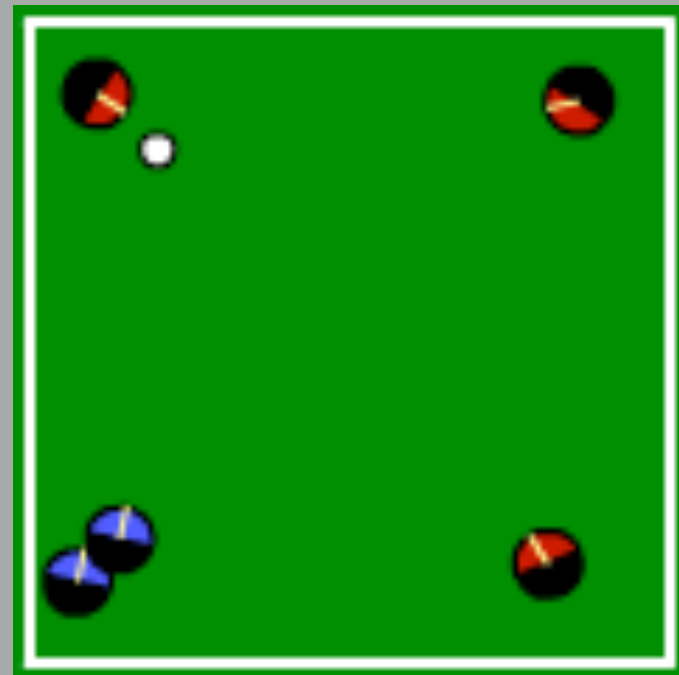


Goal
region





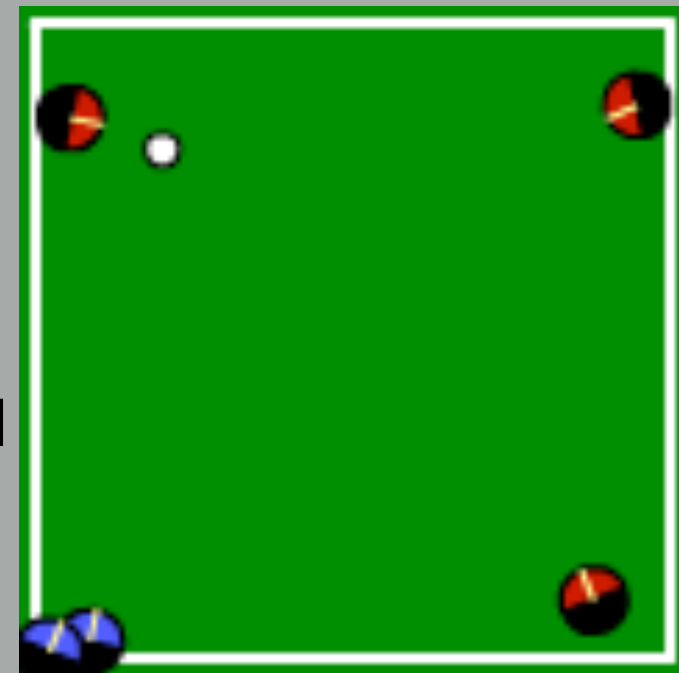
Random



Learned



Hand-coded



Hold

25



15



8



5



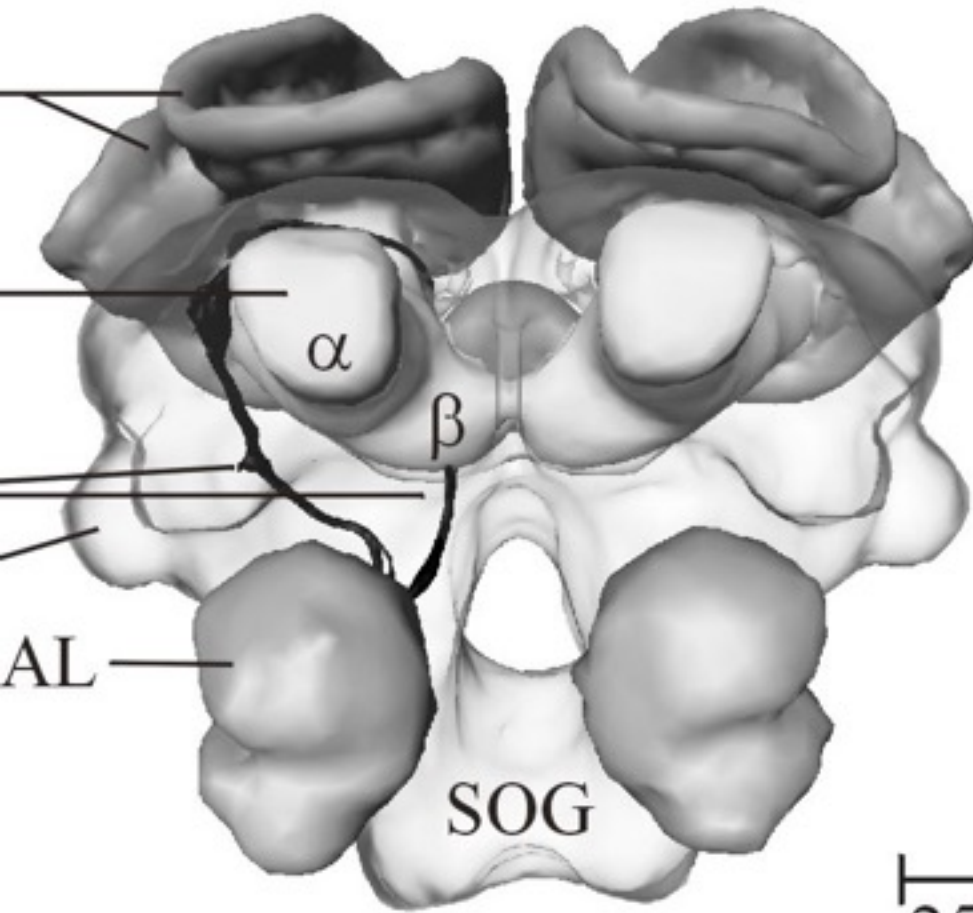
10



Temporal-difference (TD) error

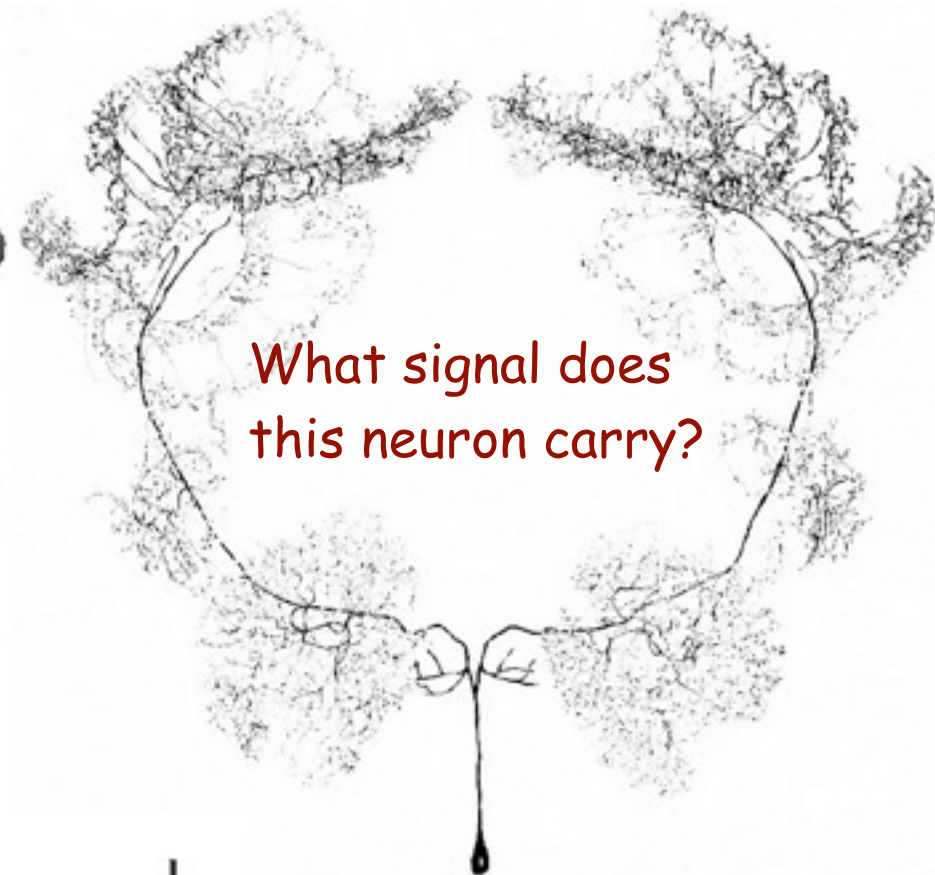
Do things seem to be getting better or worse,
in terms of long-term reward,
at this instant in time?

Brain reward systems



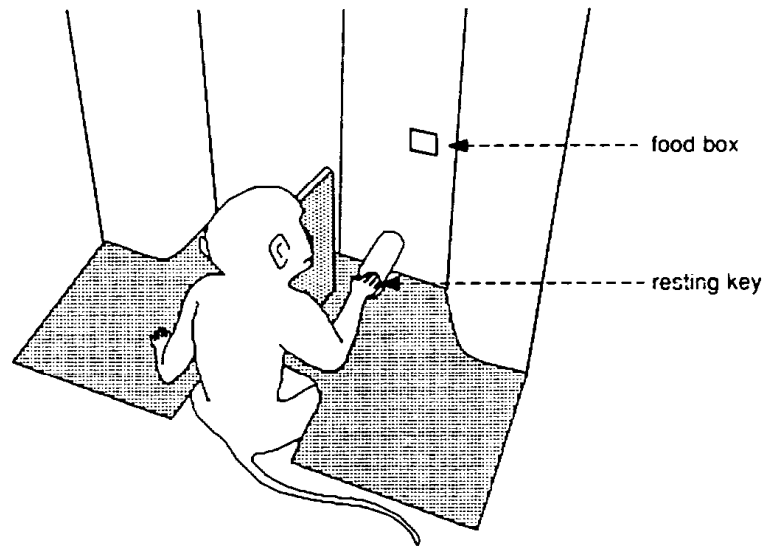
Honeybee Brain

250 μm



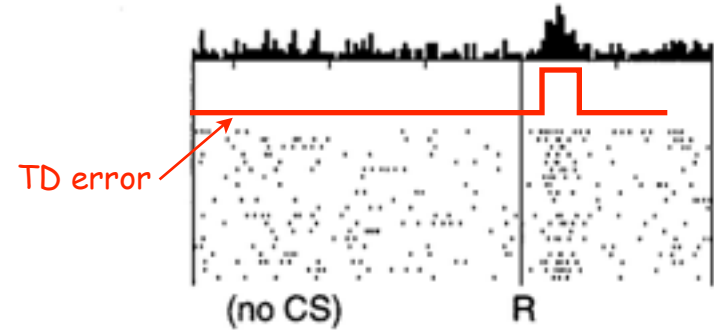
VUM Neuron

Brain reward systems seem to signal TD error

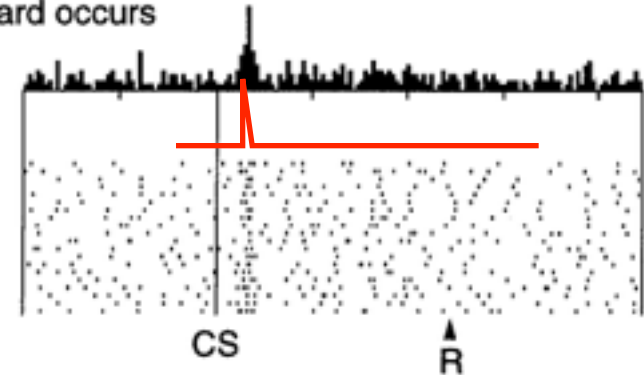


Wolfram Schultz, et al.

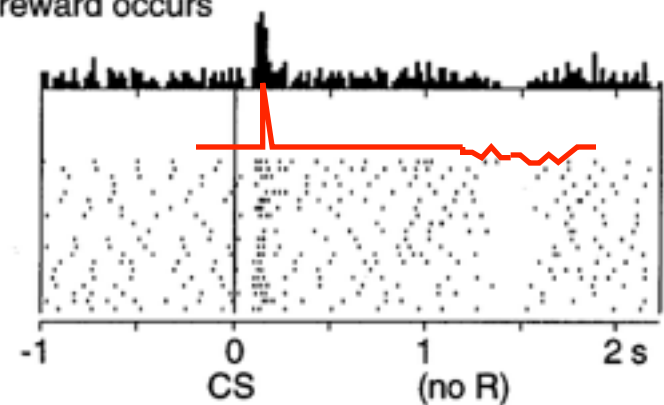
No prediction
Reward occurs



Reward predicted
Reward occurs

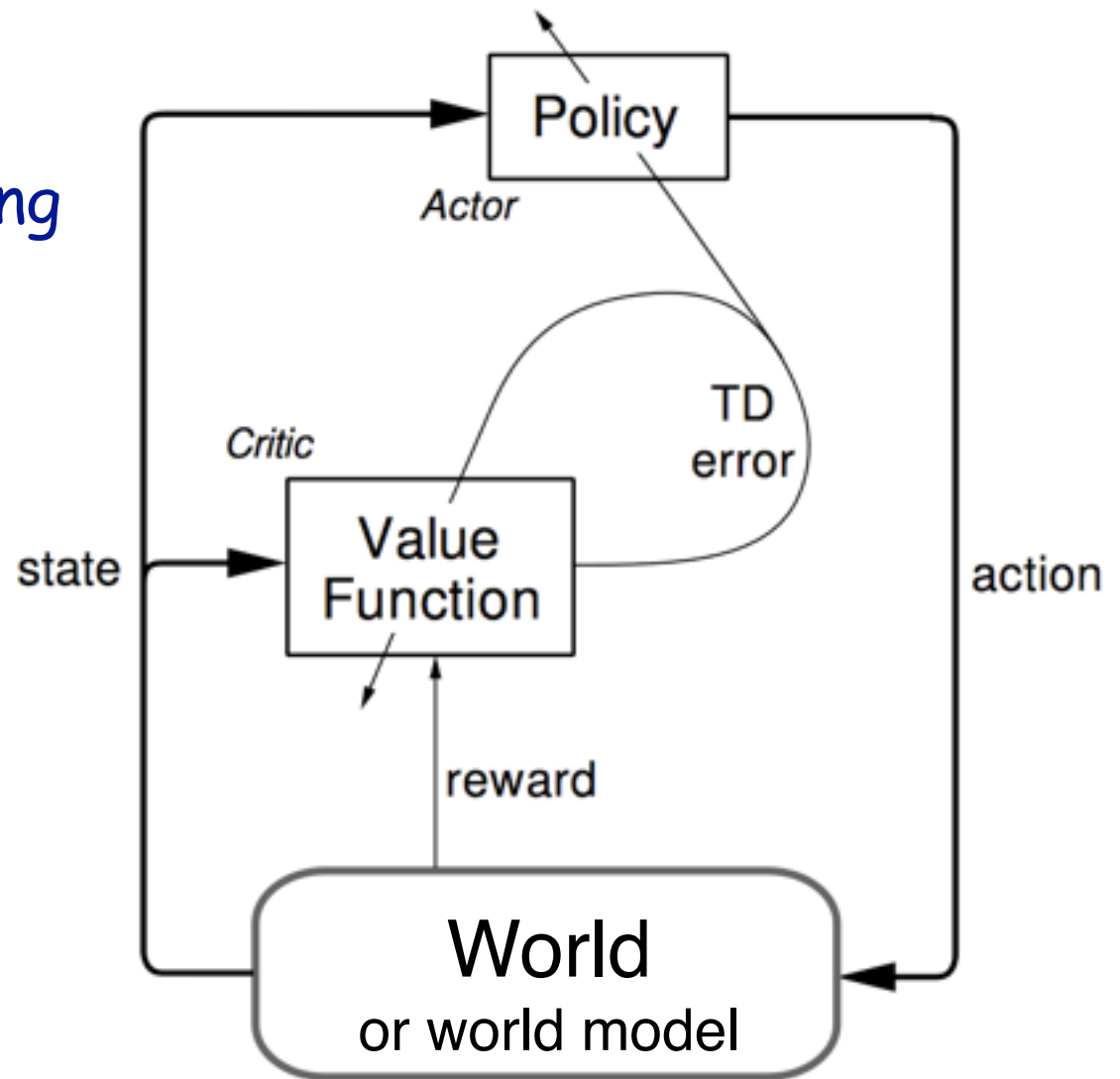


Reward predicted
No reward occurs



World models

the actor-critic
reinforcement learning
architecture



“Autonomous helicopter flight via Reinforcement Learning”

Ng (Stanford), Kim, Jordan, & Sastry (UC Berkeley) 2004

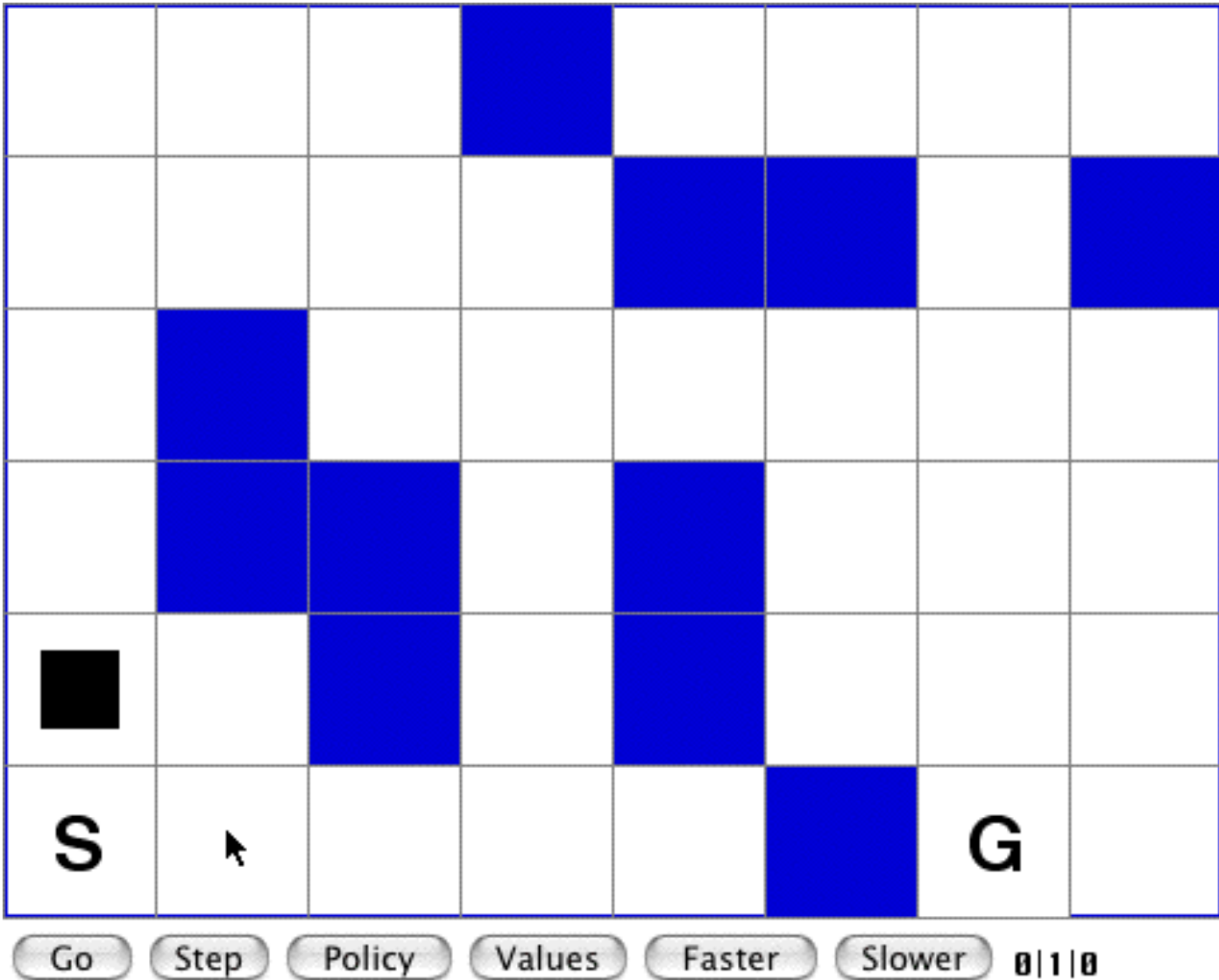




Reason as RL over Imagined Experience

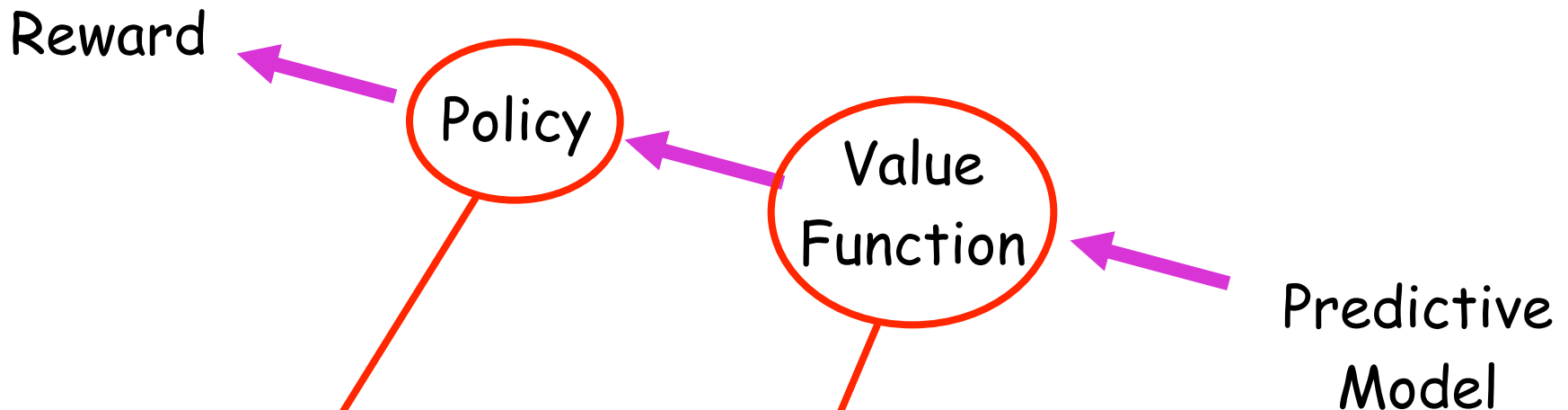
1. Learn a predictive model of the world's dynamics
transition probabilities, expected immediate rewards
2. Use model to generate imaginary experiences
internal thought trials, mental simulation (Craik, 1943)
3. Apply RL as if experience had really happened
vicarious trial and error (Tolman, 1932)

GridWorld Example



Summary:

RL's Computational Theory of Mind

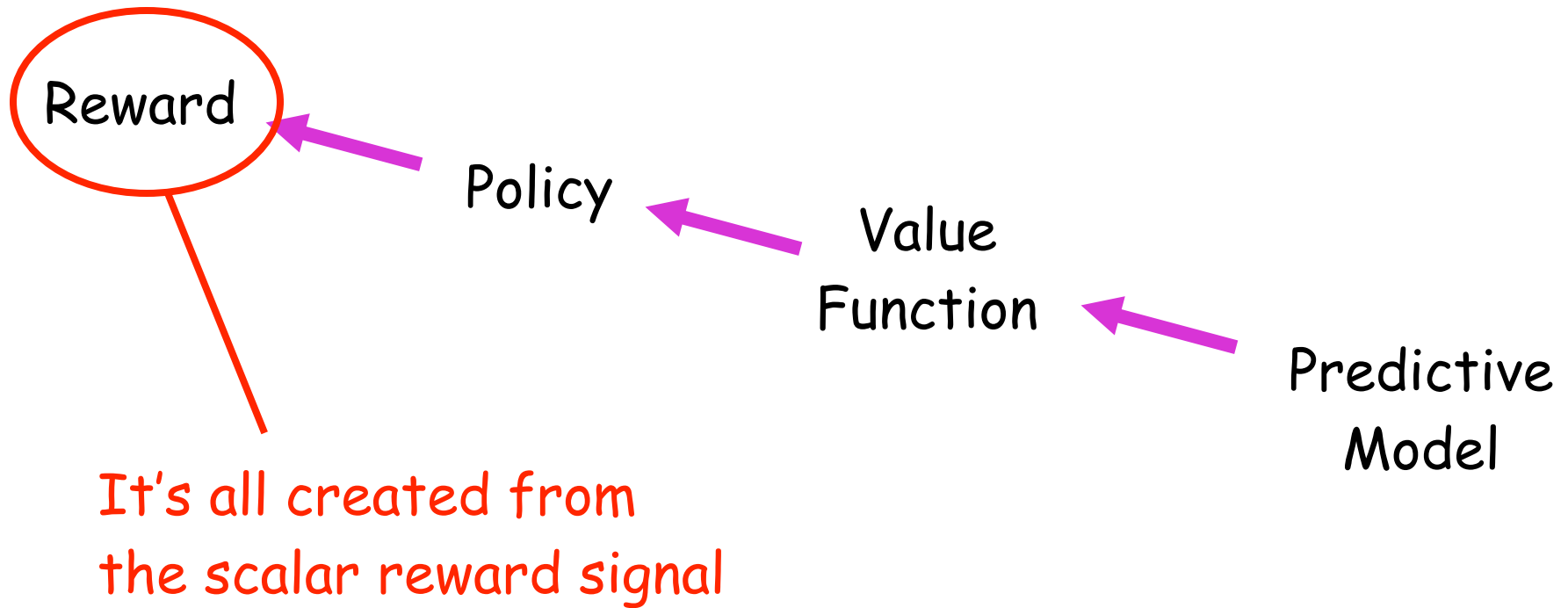


A learned, time-varying prediction of imminent reward
Key to all efficient methods for finding optimal policies

This has nothing to do with either biology or computers

Summary:

RL's Computational Theory of Mind



Summary:

RL's Computational Theory of Mind

